

# Principal Component Analysis Based Backpropagation Algorithm For Diagnosis of Peripheral Arterial Occlusive Diseases

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**Abstract--** Impedance cardio-vasography (ICVG) serves as a non-invasive screening procedure prior to invasive and expensive angiographic studies. Parameters like Blood Flow Index (BFI) and Differential Pulse Arrival Time (DPAT) at different locations in both lower limbs are computed from impedance measurements on the Impedance Cardiograph. A Backpropagation neural network is developed which uses these parameters for the diagnosis of peripheral vascular diseases such as Leriche's syndrome. The target outputs at the various locations are provided to the network with the help of a medical expert. The paper proposes the use of Principal Component Analysis (PCA) based Backpropagation network where the variance in the data is captured in the first seven principal components out of a set of fourteen features. Such a Backpropagation algorithm with three hidden layers provides the least mean squared error for the network parameters. The results demonstrated that the elimination of correlated information in the training data by way of the PCA method improved the networks estimation performance. The cases of arterial Narrowing were predicted accurately with PCA based technique than with the traditional Backpropagation Technique. The diagnostic performance of the neural network to discriminate the diseased cases from normal cases, evaluated using Receiver Operating Characteristic (ROC) analysis show a sensitivity of 95.5% and specificity of 97.36% an improvement over the performance of the conventional Backpropagation algorithm. The proposed approach is a potential tool for diagnosis and prediction for non-experts and clinicians.

## I. INTRODUCTION

Impedance Plethysmography relates to an indirect assessment of blood volume changes in any part of the body segment by measurement of its Electrical Impedance as a function of time. The Impedance cardio-vasograph (ICVG) is based on the principle of Impedance Plethysmography and provides normalized rate of change of impedance ( $dZ/dt$ ) waveforms for assessment of peripheral blood in the lower limbs [1]. Since blood is a good conductor of electricity, the amount of blood in a given body segment is reflected inversely as in the electrical impedance of the body segment. Pulsatile blood volume increase in the body segment caused by systemic blood circulation therefore, causes proportional decrease in the electrical impedance [2] [7]. The waveform is obtained by taking the natural logarithm of the instantaneous impedance of the body segment and then

differentiating with respect to time [3]. The waveform thus obtained has its amplitude proportional to the blood flow index in the body segment and free from geometric variations of the limb [4]. Non-invasive techniques like vascular Doppler, venous occlusion method and Plethysmography are able to track the obstructions in the blood flow in the arteries and veins and thus assist in the diagnosis of peripheral vascular diseases [5]. The venous occlusion principle could be used to estimate the peripheral blood flow only in the distal segments of the extremities due to application of a tourniquet in the proximal part of the limb. Thus it was possible to detect the arterial blood in the limb but it was not possible to specify the approximate anatomical location of the block. Due to application of the tourniquet the method became offline and it was not possible to assess peripheral blood flow in real time [6]. A feed forward Backpropagation network is designed consisting of an input layer, an output layer and three hidden layers. Data of about 130 lower extremities (65 patients) at four locations (thigh, knee, calf and ankle) are used for testing the network. This configuration makes the mapping easier as the first hidden layer extracts local features of the function while the global features are extracted in the second hidden layer.

## II. IMPEDANCE CARDIO-VASOGRAPHY

The evaluation of Nyober's parallel conductor theory forms the physical basis of Impedance Cardio-vasography [2].

$$dV = -\rho_b \frac{L^2}{Z^2} dZ \quad (1)$$

which relates the change in the blood volume ( $dV$ ) to the change in the impedance of the body segment ( $dZ$ );  $\rho_b$  is the resistivity of blood. For the assesment of the peripheral blood flow in a particular limb the change in impedance  $dZ$  is replaced by the product of the maximum amplitude and duration of the systolic wave[10].

$$dV = -\rho_b \frac{L^2}{Z_o^2} (dZ/dt)_m T \quad (2)$$

$Z_o$  is the basal impedance of the limb,  $T$  is period during which the systolic blood flow takes place and  $(dZ/dt)_m$  is the maximum amplitude of the  $dZ/dt$  waveform

Further  $\rho_b L^2$  is replaced by  $Z_o V$  so that equation 2 becomes

$$dV = \frac{(dZ / dt)_m T}{Z_o} \quad (3)$$

Blood Flow in ml per 1000 cc of body tissue per cardiac cycle can be thus estimated as

$$dV = 1000 \frac{(dZ / dt)_m T}{Z_o} \quad (4)$$

where division by  $Z_0$  is used to normalize the  $dZ/dt$  waveform. The  $dZ/dt$  waveform used for obtaining the parameters BFI and DPAT for the neural network is preferred because: (a) The phase reversal points are clearly detectable from the waveform and represent important events of the cardiac cycle, (b) The blood flow related signals are enhanced and respiratory component is suppressed due to differentiation., (c) the maximum rate of change of impedance  $(dZ/dt)_m$  can be directly measured as the amplitude of the positive going peak and therefore the human error involved in the extrapolation of the forward slope in impedance waveform is avoided. The maximum amplitude of the normalized  $dZ/dt$  waveform is represented by Blood Flow Index (BFI). The time elapsed from ventricular contraction to the highest point on the obtained waveform is known as the Pulse Arrival Time (PAT). It was observed that BFI and PAT changed significantly in patients with Peripheral Arterial Occlusive Diseases as compared to Normal subjects. The parameter DPAT is obtained by subtracting the PAT

About two hundred patients in the age group of 10 to 75 years, who presented at various surgical OPD's of K.E.M hospital and J.J. Hospital at Mumbai, India, were subjected to this study. The Table 1 gives the representative data for the subject (Indoor/OPD Reg No 666)

Table 1: Parameters for subject (Indoor/OPD Reg No 666)

| Location | Left Leg |      | Right Leg |      |
|----------|----------|------|-----------|------|
|          | BFI      | DPAT | BFI       | DPAT |
| THIGH    | 0.20     | 00   | 0.43      | 30   |
| KNEE     | 0.51     | 160  | 0.63      | 30   |
| CALF     | 0.39     | 00   | 0.55      | 00   |
| ANKLE    | 0.31     | 00   | 0.88      | 40   |

The above parameters along with the target outputs were given to the neural network for training. Normal volunteers were used to provide accurate identification of  $dZ/dt$  peaks among various pathological cardiovascular states. The network is designed with an input layer consisting of fourteen neurons which correspond to the BFI and DPAT values as shown in Table 1. Because of the interdependence of parameters (correlation of 0.9), data of both the limbs is applied simultaneously as the input to the net during the training as well as the testing process. The output layer consists of 8 neurons, two for each location. Basically the output is first classified as Non Block or Block at a particular location, if it is Non block then it is further classified as Normal or Narrowing and if it is a Block it checks the status of collateral circulation and also for a further Block at the ankle. The outputs are summarized in Table 2.

Table 2(a): Targets in case of Block

| Block at a particular Location (11) then subsequent locations |    |
|---------------------------------------------------------------|----|
| Good Collaterals                                              | 00 |
| Moderate Collaterals                                          | 01 |
| Poor Collaterals                                              | 10 |

|                       |    |
|-----------------------|----|
| Further Block (ankle) | 11 |
|-----------------------|----|

Table 2(b): Targets in case of Non-Block

| Non-Block at a particular Location (00) then subsequent locations |    |
|-------------------------------------------------------------------|----|
| Normal                                                            | 00 |
| Narrowing                                                         | 10 |

However instead of '0' and '1' as outputs we give the targets as '-1' and '1', so that a threshold of 'zero' decides the decision boundary.

### III. PCA TECHNIQUE

The Backpropagation network is developed with three hidden layers each layer consisting of twelve neurons. The figure of 12 hidden neurons was decided as the average value of the input parameters (16) and the output layer neurons (8). The performance of this network was evaluated against Backpropagation networks having one and two hidden layers as shown in Table 3. Each of the networks is trained for a goal of 0.01. In the second part of the analysis the PCA technique is applied before applying the above developed Backpropagation network. For implementing the PCA technique it is necessary to standardize the data which is done by converting it to data with zero mean. The PCA projects data along directions where data varies the most. These directions are determined by the eigen value of covariance matrix corresponding to the largest eigen value [8] [9]. Each principal component is a linear combination of the original variables. The principal components as a whole form an orthogonal basis for the space of the data. The first principal component is a single axis in space. When we project each observation on that axis, the resulting values form a new variable. And the variance of this variable is the maximum among all possible choices of the first axis. The second principal component is another axis in space, perpendicular to the first. Projecting the observations on this axis generates another new variable. The variance of this variable is the maximum among all possible choices of this second axis. The full set of principal components is as large as the original set of variables. The following procedure is followed to accomplish the above: (i) standardize the data by dividing each column by its standard deviation. (ii) Determine the Coefficients of the four Principal Components. These coefficients are the linear combinations of original variables that generate new variables, they represent eigen vectors of the matrix  $x^T x$ , where  $x$  is the data matrix of size  $(n \times p)$ . Columns of Coefficient Matrix are in decreasing order of component variance. The coefficient matrix is of size  $(p \times p)$ . (iii) The Principal Components are obtained by product of the data matrix with the coefficient matrix. (iv) The variance vector is calculated which gives the variances explained by each Principal Component. The variances represent the eigen values of the data matrix and they are ordered in the decreasing order of magnitude. (v) The percent of total variability contributed by each Principal Component can be calculated as

$$\%Variance = 100 \frac{\text{variances}}{\text{sum}(\text{variances})} \quad (5)$$

This Percentage variation is shown as a Pareto Chart in Figure 1. The figure shows that the only clear break in the amount of variance accounted for by each component is between the first and second components. However, that component by itself explains about 30% of the variance, so more components are needed. It is observed from the figure that the first seven principal components explain roughly ninety five percent of the total variability in the standardized ratings, so that is a reasonable way to reduce the dimensions in order to visualize the data. The PCA procedure eliminates those principal components that contribute less than 5% to the total variation in the data set as seen from Figure 1.

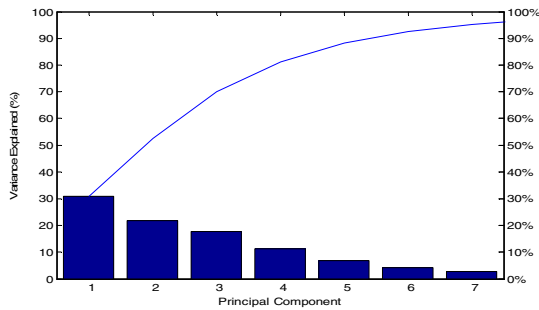


Figure 1: Percent Variability contributed by each Principal Component.

#### IV SIMULATIONS & RESULTS

The mean square error (mse) for the Backpropagation and PCA based networks methods are shown in Table 3.

Table (3): Comparison of the conventional backpropagation and PCA based networks for learning rate=0.01.

| Hidden layers    | Backpropagation |       | PCA based Backpropagation |       |
|------------------|-----------------|-------|---------------------------|-------|
|                  | mse             | Epoch | mse                       | Epoch |
| Three 12 neurons | 0.0099993       | 10000 | 0.00998835                | 4997  |
| Two 12 neurons   | 0.0189853       | 10000 | 0.00999853                | 5601  |
| One 12 neurons   | 0.0461207       | 10000 | 0.00999985                | 9998  |

The performance of PCA based Backpropagation network is shown in Figure 2.

The ROC analysis of the network with PCA is carried out and the results are as shown in the Table (4). A similar table (5) shows the ROC analysis of the conventional Backpropagation Network. Based on the ROC analysis the Table (6) compares the performance of the above two networks.

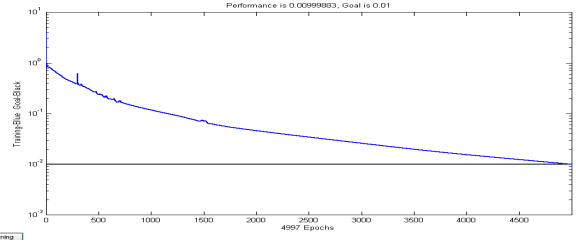


Figure 2: Performance of PCA based Backpropagation Network

Table (6): Performance Comparison of the two networks

| Parameters                | Back propagation | PCA based Back propagation |
|---------------------------|------------------|----------------------------|
| Sensitivity               | 93.1%            | 95.5%                      |
| Specificity               | 87.8%            | 97.36%                     |
| Positive likelihood ratio | 7.6311           | 36.31                      |
| Negative likelihood ratio | 0.07858          | 0.0456                     |
| Positive predictive value | 94.19%           | 98.85%                     |
| Negative predictive value | 85.71%           | 90.24%                     |

From the above table we can see that there is an improvement in the Sensitivity ( $TP/TP+FN$ ) and Specificity ( $TN/(TN+FP)$ ) of PCA based Backpropagation Networks. The Positive Likelihood ratio (Sensitivity/ (1-Specificity)) for PCA network gives a conclusive increase in the probability of disease when the test is positive. Furthermore the Positive ( $TP/(TP+FP)$ ) and Negative Predictive values are also sufficiently higher for the PCA base network.

#### V CONCLUSIONS

The sensitivity (95.5%) and specificity (97.36%) of the Neural Network for patient data which are not a part of the training set as shown in the ROC Analysis is an improvement over that obtained by Karamchandani et al [1]. The PCA technique enhances the performance of the network as it orthogonalizes the components of the input vectors (so that they are uncorrelated with each other), it orders the resulting orthogonal components (principal components) so that those with the largest variation come first, and eliminates those components that contribute the least to the variation in the data set[11].

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Table (4): ROC Analysis of PCA based Backpropagation Network

| Disease         |                     |                   |                     |                   |                   |
|-----------------|---------------------|-------------------|---------------------|-------------------|-------------------|
| Test            | Present             | n                 | Absent              | n                 | Total             |
| <b>Positive</b> | True Positive (TP)  | <b>86 (a)</b>     | False Positive (FP) | <b>1 (c)</b>      | <b>87 (a + c)</b> |
| <b>Negative</b> | False Negative (FN) | <b>4 (b)</b>      | True Negative (TN)  | <b>37 (d)</b>     | <b>41 (b + d)</b> |
| <b>Total</b>    | --                  | <b>90 (a + b)</b> | --                  | <b>38 (c + d)</b> | --                |

Table (5): ROC Analysis of PCA based Backpropagation Network

| Disease         |                     |                   |                     |                   |                   |
|-----------------|---------------------|-------------------|---------------------|-------------------|-------------------|
| Test            | Present             | n                 | Absent              | n                 | Total             |
| <b>Positive</b> | True Positive (TP)  | <b>81 (a)</b>     | False Positive (FP) | <b>5 (c)</b>      | <b>86 (a + c)</b> |
| <b>Negative</b> | False Negative (FN) | <b>6 (b)</b>      | True Negative (TN)  | <b>36 (d)</b>     | <b>41 (b + d)</b> |
| <b>Total</b>    | --                  | <b>87 (a + b)</b> | --                  | <b>41 (c + d)</b> | --                |